inherited-f4-next and fes-f2 end to end (2026-11-12)

`inherited-f4-next`: the same counted work with faster kernels, and `fes-f2` end to end (2026-11-12)

Owner's ask (2026-11-12): "Speed it all up". The M4GB side is `../m4gb_update_signatures_20261112/`. This study covers the in-house engines that the algebraic index-calculus oracle actually runs.

Correction to the brief

The change was briefed as "`inherited-f4` is the best engine on every cell". That holds for end-to-end S in the e2e ladder (`../m4gb_e2e_ic_ladder_20261110/`), which prices `inherited-f4` from its word-XOR count, and in that ladder `fes-f2` was not an arm. On wall time per call, `fes-f2` (Gray-code exhaustive search) is below `inherited-f4` up to 24 unknowns, here and already in the descent ladder (`../m4gb_descent_ladder_20261110/RESULTS.md`, R 21: 3.9 against 9.5 ms) `[pcc:unverified]`. `fes-f2` is also priced in word XORs (two per Gray-code point), so its S is on the same basis as `inherited-f4`'s; at 14 unknowns its count per call is below `inherited-f4`'s (about 33 thousand against 49 thousand in the profiling runs), from 20 unknowns up it is 4 to 13× above. The end-to-end run below measures what that does to S.

Profile (Callgrind, `inherited-f4`, AVX2 path)

`gf2_elim` chooses AVX-512 at run time on this host; Valgrind cannot decode it, so the profile is of the AVX2 path (`KIC_GF2_FORCE_AVX2=1`), instruction counts with cold caches `[pcc:unverified]`:

inclusive share	**K 23:12:2**	**R 21:11:2**	**K 13:7:2**
tree specialisation (`specialise_shared`)	64 %	59 %	55 %
↳ eager rewrite of kept rows (`apply_step_bmi2`)	20 %	16 %	11 %

inclusive share	K 23:12:2	R 21:11:2	K 13:7:2
↳ re-reduction of displaced rows (`ReducedBasis::insert`)	16 %	13 %	14 %
root (`reduce_inherited`)	22 %	29 %	31 %
↳ Macaulay build	12 %	19 %	20 %
↳ root echelon (`eliminate_core`)	9 %	9 %	9 %
system substitution	9 %	5 %	5 %

The structure-reuse-across-targets question (“dynamic programming”): the column layout is already cached across targets (keyed by the occurring variables and the degree); what a root still does per target is numeric, and 55 to 64 % of the cost is the specialisation down the tree, which is counted work. A complete-layout formula index was tried and gave nothing measurable; the existing Gröbner-trace option (`KIC_F4_ROOT_REPLAY`) drops rows and can change the tree, so it does not meet the identical-decisions rule.

What changed (engine `inherited-f4-next`; `inherited-f4` unchanged)

A per-thread switch (`koblitz_groebner::with_next_kernels`) selects, for the same eliminations and rewrites:

1. SIMD row XOR in `ReducedBasis::insert` (AVX-512 or AVX2 at run time);
2. the `bitop/v1` tally counted inside the rewrite pass instead of a second pass over the same words (identical tally);
3. Four Russians table reuse (`gf2_elim::set_reuse_table_thread`, the exact `KIC_GF2_REUSE_TABLE` behaviour);
4. existing exact matrix builders that were off by default (direct fused pack, support-local streaming and bitmap columns), whose tests assert identical matrices;
5. environment variables read once instead of per node.

None of these changes the engine’s `ops` (word XORs): the same work, done cheaper. Tests: `koblitz_groebner::tests::next_kernels_leave_decisions_and_word_xors_unchanged` (33 fixed systems, cold and warm, identical roots, tree statistics and word count), `gf2_elim::tests::reuse_table_override_leaves_rows_and_count_unchanged`, `inherited_f4::tests::simd_row_xor_matches_the_scalar_one`; module suites `koblitz_groebner` (92), `gf2_elim` (12), `inherited_f4` (14), `ic_framework::solvers` (13), `pq_f4_f2` (13) pass. Written by a subagent to a frozen brief; reviewed in the main session.

Run 1: paired per call (`run_ab.sh`)

`ic descent --solver inherited-f4 --solver inherited-f4-next --solver fes-f2`, Koblitz and random cells, m = 2 at 14 to 24 unknowns plus Koblitz 30 and 32 (degrees 25 and 27 are composite and have no Koblitz cell), 6 seeded targets × 3 repeats,

60 s per call, one thread; engine order rotates per repetition. Record `20261112T0630Z-inherited-f4-next-ab`.

Expectation: next/old 0.85 to 0.95 per call on every cell, `ops` equal on every target, 0 disagreements; `fes-f2` below both on wall up to 24 unknowns and above them at 30 to 32. Falsifier: any `ops` difference or disagreement, or a median paired ratio at or above 1.

Run 2: end to end (`run_e2e.sh`)

`ic bench --sweep` on the six instances of the e2e ladder (Koblitz 13, 17, 19, 23; random 15, 21; same subspace bases, 2 repeats, 60 s per call, 16 matched rho runs), the algebraic oracle with `inherited-f4`, `inherited-f4-next` and `fes-f2`, the pair table as reference row.

Expectation, stated before the run: `inherited-f4` and `inherited-f4-next` have the same S on every instance (same count, pinned pricing) and the second has the lower wall; `fes-f2` has a lower S than `inherited-f4` at degree 13 (14 unknowns) and a higher S from degree 17 or 19 up. If the first holds, it is a measured gain `[pcc:unverified]` on the best algebraic row at degree 13 and nothing else. Falsifier: `inherited-f4` and `-next` S differing beyond run noise, or `fes-f2` S at degree 13 not below `inherited-f4`'s.

Results

Filled after the runs.

Median over targets of the per-target median wall (ms per call, one thread). Paired: median (min–max) over targets of the per-target ratio. `ops` = median per-target word-XOR count (the engines' counted unit). `dec` = targets decided / targets; every decided answer is checked against the reference `fes-f2`.

family	n	vars	inherited-f4 ms	-next ms	next/old paired	targets next < old	ops equal	fes-f2 ms	fes/old
K	13	14	1.03	0.94	0.918 (0.90–1.00)	6/6	yes	0.09	0.09
R	13	14	1.19	1.13	0.926 (0.87–0.95)	6/6	yes	0.10	0.08
K	15	16	2.40	2.22	0.930 (0.83–0.95)	6/6	yes	0.19	0.08
R	15	16	1.30	1.18	0.914 (0.90–0.93)	6/6	yes	0.12	0.10
K	17	18	2.21	2.08	0.941 (0.91–1.02)	5/6	yes	0.24	0.11

family	n	vars	inherited-f4 ms	-next ms	next/old paired	targets next < old	ops equal	fes-f2 ms	fes/old
R	17	18	2.22	2.04	0.920 (0.89–0.97)	6/6	yes	0.25	0.11
K	19	20	4.25	3.65	0.860 (0.81–0.86)	6/6	yes	0.82	0.21
R	19	20	5.58	5.11	0.893 (0.84–0.98)	6/6	yes	1.24	0.21
R	21	22	7.23	6.38	0.882 (0.84–0.89)	6/6	yes	2.77	0.36
K	23	24	15.74	13.71	0.886 (0.81–0.90)	6/6	yes	10.73	0.69
R	23	24	14.31	12.47	0.894 (0.85–1.28)	5/6	yes	11.34	0.80
K	29	30	90.47	81.70	0.893 (0.80–0.95)	6/6	yes	705	7.75
K	31	32	217	192	0.872 (0.80–1.15)	4/6	yes	3,036	14.22